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Professor Sparks

Stats 20

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Final Project Report

For our final project, we analyzed a large, real-world dataset from the California Safe Cosmetics Program, which tracks cosmetic products containing chemicals linked to potential health risks. The approved dataset, obtained from data.gov, includes 16,555 observations and 21 variables, covering product names, categories, reporting dates, and more. Our goal was to analyze patterns in chemical usage across cosmetic products. We decided to divide the work evenly so everyone got some experience with each part of the project. Each of us made two visualizations, ran three statistical tests, and cleaned our own data. We then worked together to write the report and build the slide deck.

Because our data was publicly sourced from an online database, we needed to clean and organize the variables to make the dataset workable. This involved formatting date columns to extract usable information, handling missing text entries, and filtering observations with incomplete chemical data. We also found some items with a chemical count of zero, despite the dataset primarily focusing on products containing chemicals. Since most of our analyses dealt with examining trends in products with chemicals, products without chemicals were filtered out for the most part to avoid bias in those analyses.

After the cleaning, we went ahead and graphed out some of the variables to see what trends we could analyze further. The initial graphs included the number of products reported per year as well as the top reported chemicals in the products. We then examined deeper patterns by

graphing the top product types, as well as the top brands, followed by a breakdown of subcategories to determine which contributed the most to the dataset. We also checked for discontinued products, but since a majority hadn't been discontinued, we decided that we would not investigate that variable in many of our further analyses.

Once we got a general idea of the data, we began statistical testing to answer more specific questions that we found interesting. Carly used a two sample t-test, goodness of fit chi-square test, and a correlation test to examine the data. The correlation test looked at whether newer products tended to have more chemicals. The result showed a weak positive relationship ($r \approx 0.13$), meaning newer products might actually have slightly higher chemical counts. Brooke utilized linear regression, one-sample t-test, and logistic regression. The logistic regression analysis was used to see if the presence of chemicals in products had an influence on them being discontinued. The result revealed that products without chemicals had about a 24.6% chance of being discontinued while products with chemicals only had a 11.2% chance of being discontinued. This was surprising because revealed products without chemicals in them being more likely to be discontinued compared to products with chemicals in them. Vivian used a chi-square test for independence, ANOVA, and 2 prop z-test. Based on the test for independence, we found that certain brands focus more on specific types of categories. For example, Sephora primarily focuses on makeup products. From our earlier visualizations, we saw that makeup products contain many of the most frequently reported chemicals. This helps explain why Sephora appears among the top brands associated with high chemical counts as makeup is the category in which these chemicals are most common. Thus, makeup-focused brands naturally appear more frequently in chemical reports.

One of our biggest challenges was figuring out how to work with such a large dataset. With over 16,500 observations, a lot of our visuals ended up looking crowded unless we left out data, which wasn't ideal. We also ran into situations where a graph looked "wrong" even though the code wasn't. At the end, we found that some values were stacked on top of each other or clustered in weird ways and we had to jitter points to get the plots to look readable. This helped us learn how messy real data can be and how much trial and error goes into getting visuals to look right.

If we continue this project, we'd want to look into more information such as how specific companies or brands use certain chemicals and whether there are any trends in how they change over time. Examining the prices and ratings of these products in relation to the presence of chemicals would also tell us how consumerism influences product formulation. It would also be interesting to look at the health impacts of the chemicals themselves, not just how often they show up. If this project were expanded, we would be able to explore further safety concerns and possibly connect the dataset with outside sources to see the larger impact of the chemical use.

AI Usage Statement

We affirm that generative AI tools (including but not limited to ChatGPT, Copilot, Gemini, and Claude) were not used at any stage of this project. This includes idea generation, code writing, debugging, documentation, or any other form of assistance. All work presented here is the original product of our group members, developed through our own reasoning, collaboration, and programming efforts.